Weekly Report

Downstate Grid Refinement and 2025 Operating Inputs

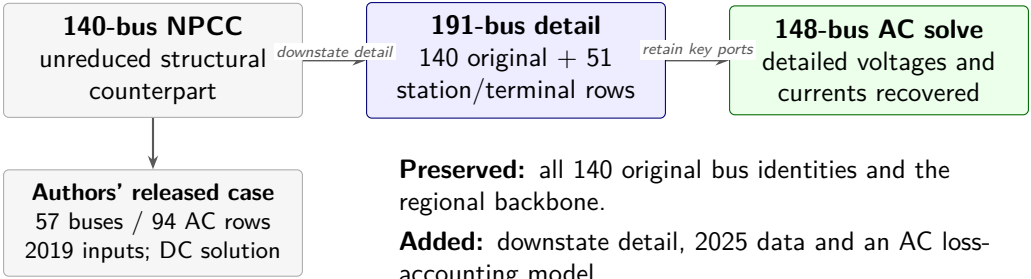
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The 2019 Reference and the Current Model

Liu et al., IEEE TPWRS 2023; 2019 is the example operating-data year



Preserved: all 140 original bus identities and the regional backbone.

Added: downstate detail, 2025 data and an AC loss-accounting model.

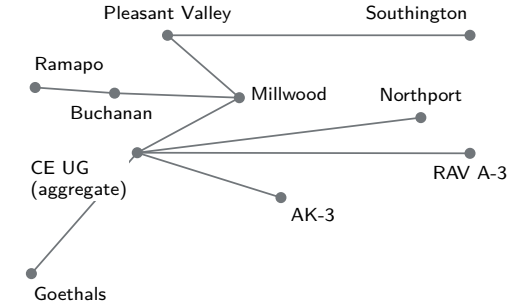
The current expanded network has **266/267 buses**, including numerical line-section points—not 267 substations.

Source: [1] Author-release audit; [2] downstate/slim construction; [4] current model inventory.

Downstate Network Modifications

2019 structural counterpart

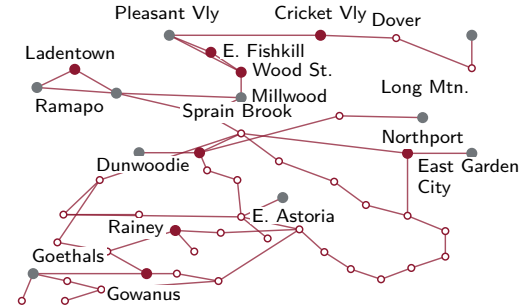
Paper-derived NPCC parent; G–K excerpt



Schematic; selected network excerpt.

Current downstate detail

Station detail + dated projects



Eight added 345-kV ports remain explicit; **43 other added terminals are reduced and recovered**.

Source: [2] Original NPCC and downstate branch registers. Selected G–K connectivity; station voltage ports and parallel circuits combined for display.

Corridor Detail and Dated Operating States

Area / project	What is represented	Timing used in 2025
Hudson–Westchester	East Fishkill, Wood Street and Ladentown split bulk paths	Historical station detail
Rainey–Corona (Queens)	Cable, transformer and PAR	Present all year; completed May 2023
Gowanus–Greenwood (Brooklyn)	Receiving cable and RCC controls	May 21, 00:00 EDT
Goethals–Fox Hills (Staten Island)	Cable / transformer / PAR	May 30, 00:00 EDT
Cricket Valley–Dover corridor	Two parallel Dover PAR banks; Long Mountain connection	Full service: Dec. 19, HB23 EST

- Nine predecessor aggregates are deactivated to avoid duplicating paths.
- F84 energized **July 28, 2019**; most Long Island additions expose historical detail.

Source: [2,3] Source/upgrade registers, owner notices and Dover audit. Day-precision May completion dates use the next midnight; installed electrical parameters remain partly assumed.

Updating the Operating Inputs to 2025

Input	Public source and resolution	Model treatment
Zonal demand	NYISO P58C: hourly, 11 zones	Exact gross-load totals; assumed bus shares
Generation	P63 fuel mix; EIA-860 inventory; EIA-923 plant energy	Bounded unit allocation; plant dispatch remains reconstructed
Interchange	EIA-930 bilateral flows: hourly	Actual-report data; hour-ending converted to hour-start
PAR controls	NYISO P34: nominal five minutes	Hourly aggregation; sign, meter and gap assumptions retained
Evaluation only	NYISO P32 interface series	Not used to fit input values or repair parameters

One calendar: 8,760 physical hours, explicit DST handling, **109 source exclusions**.

Reconstructed dispatch: reconcile load, generation and interchange before AC solution.

Source: [3,4] Input, regional-boundary and PAR receipts. P53A schedules are separate evidence, not a substitute for P34 reported controls.

AC Reconstruction of the Declared Network

- Replace the DC-only operating calculation by coupled AC power flow with explicit loss allocation.
- Retain **148 voltage buses**; recover all detailed voltages and branch currents through the complex-admittance reduction.
- Represent overhead sections, cables, transformers and PARs separately. Conductor DLR excludes voltage-changing branches.

Accounting identity

$$P_z^{\text{gross}} = P_z^{\text{served}} + L_z^{\text{allocated}}$$

Fixed native generation and regional targets are preserved during loss balancing.

Engineering corrections

Parent 41: coherent R/X/B and conductor realization.

Parents 185/195: distribute the same nominal reactor compensation to section endpoints.

Source circuits, equivalent connections and engineering R/X/B assumptions remain separately identified. No claim of a complete installed-asset reconstruction.

Source: [4,5] Current model structure; physical-repair and independent raw-physics receipts.

Annual Validation and Downstate Performance

Latest completed 2025 annual replay

8,651 / 8,651 eligible hours
pass equations, loss accounting and recovery.

Zero solver exceptions; 109 excluded calendar positions stay excluded.

Parent 185/195 voltage violations remain **zero** on the full eligible cohort.

Current downstate interfaces

Interface	MAE (MW)	WAPE
UPNY–ConEd	235.2	**7.84%**
Sprain–Dunwoodie	177.2	**9.59%**

These are aggregate diagnostic comparisons. Lower interface error does not validate every internal branch, bus load or physical meter.

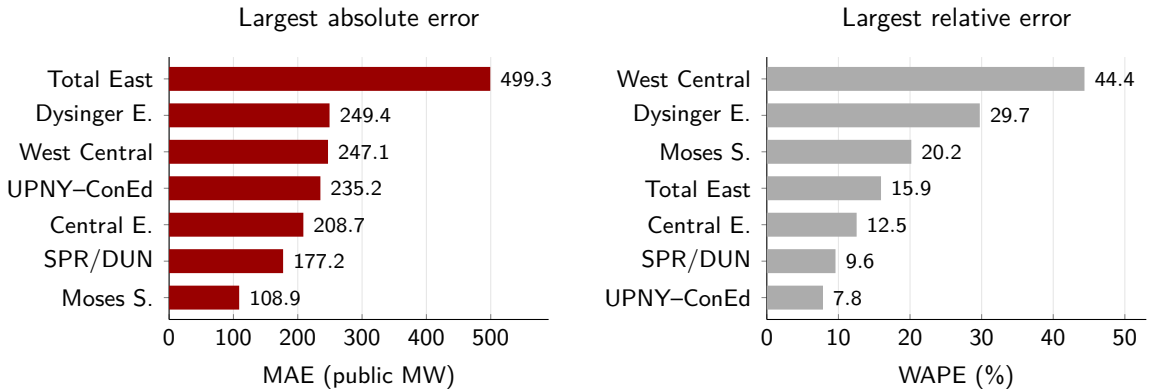
Operating qualification is still open

0 hours pass all network operating limits. Other voltage, generator Q/S and branch-rating violations remain.

Source: [4] `moses_repair_v1`; primary signed-from operator, public MW, common 8,651-hour cohort.

Where the Largest Interface Errors Occur

Current 2025 model; same 8,651 hours and frozen diagnostic definitions



Source: [4] Current annual metrics. MAE and WAPE answer different questions; the physical P32 measurement correspondence remains uncertified.

2019 Benchmark vs. 2025: Errors and Observed Flows

Released 2019 DC case (8,760 h) → current 2025 AC case (8,651 eligible h)

	MAE (MW) 2019 → 2025	Δ MW	WAPE (%) 2019 → 2025	Δ pp	NYISO mean $\lvert F\rvert$ (MW) 2019 → 2025	Δ%
Total East	286.5→499.3	+212.8	7.24→15.95	+8.71	3,960→3,131	−20.9
Dysinger East	209.9→249.4	+39.5	22.49→29.74	+7.25	933→839	−10.1
West Central	218.5→247.1	+28.6	46.81→44.36	−2.44	467→557	+19.3
UPNY–ConEd	241.1→235.2	−5.9	9.91→7.84	−2.07	2,433→2,999	+23.3
Central East	211.8→208.7	−3.1	9.63→12.54	+2.91	2,199→1,664	−24.4
SPR/DUN–South	299.6→177.2	−122.3	10.89→9.59	−1.29	2,752→1,848	−32.8
Moses South	152.7→108.9	−43.8	8.99→20.18	+11.19	1,698→540	−68.2

Δ = 2025 minus 2019; pp = percentage points. Mean $\lvert F\rvert$ is the absolute reported interface-flow magnitude.

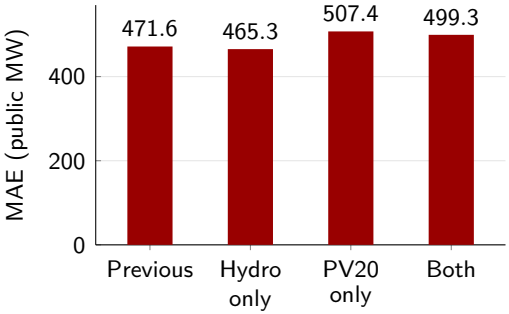
Moses South: MAE falls 28.7%, while WAPE rises 11.19 pp as mean $\lvert F\rvert$ falls 68.2%.

Reported signed mean: $+1,697 \rightarrow -190$ MW; negative-flow hours: 0.1%→68.6%.

Cross-year scope: inputs, physics, cohorts and meter definitions differ; changes do not isolate a topology effect.

Source: [1,4] 2019: author-release if.map; 2025: moses_repair_v1. Recomputed WAPE; the paper's limit-normalized metric differs. Physical meter continuity remains unverified.

Total East: Why the Latest Error Increased

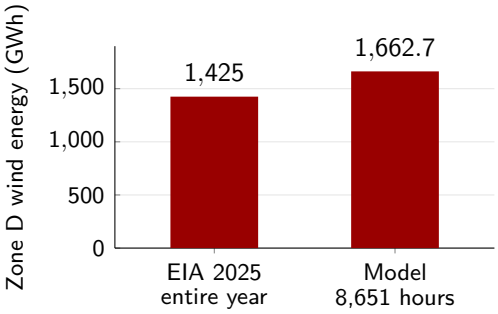


- PV20 rerouting reduces the main eastbound modeled flow, deepening an existing Total East underprediction.
- Order-averaged MAE effects: **+34.9 MW from PV20**, offset by −7.2 **MW from hydro**.
- The current **five-parent proxy** has no certified equivalence to NYISO's **29-entry physical definition**.

Two additional full-year runs isolate effects *inside this model*. Adding one physical term does not certify the complete measurement operator.

Source: [6] Controlled B/G/P/GP annual ablations; [7] NYISO operating-study definitions. All scenarios use the same 8,651 hours.

Moses South: Remaining Generation Issues



Wind: an input inconsistency

The six-plant modeled subset exceeds full-year EIA energy. A shared statewide capacity factor misses local production.

St. Lawrence: hourly uncertainty

Monthly energy matches EIA. Alternative hourly shapes change the residual; plant-hour telemetry is unavailable.

Boundary qualification

Sand Bar proxies Plattsburgh; tap load, losses and P32 mapping remain unresolved.

Under current nonnegative excluded-hour wind bounds, excess is **237.7 GWh**, or at least **27.48 MW** on qualified hours. Current Moses bias is +24.32 MW; exact causal attribution is not established.

Source: [6] Final EIA-923 six-plant audit, current profiles and controlled shape checks. EIA net output can be negative; the bound is conditional.

Western Interfaces: Source and Control Gaps

What is established

- West Central: **44.36% WAPE**; Dysinger East: **29.74%**.
- An earlier source audit matched all **108 Niagara/Lewiston/Gilboa monthly target values** to final EIA data.
- That check excludes a source-vintage/arithmetic defect; it does not validate hourly dispatch.

What remains unresolved

- Niagara's actual hourly split among three electrical terminals; its aggregate model landing remains a proxy.
- The documented Dysinger–East Stolle Road PAR path and its complete control representation.
- Physical interface membership, meter terminals and source-to-model correspondence.

Interpretation: the large residuals identify a modeling priority, not a uniquely diagnosed line-impedance error.

Source: [3,6] Western source audit and current annual metrics. Historical DC correlation results are not presented as fresh current-AC evidence.

Next Modeling Priorities

1. **Correct plant-level generation allocation.** Anchor northern wind energy to plant data while preserving statewide totals; distinguish reported monthly values from assumed hourly shapes.

2. **Complete the interface measurement operators.** Resolve Total East/Central East and western membership, terminal signs and dated controls before using errors to tune network parameters.

3. **Revalidate the operating model.** Preserve the downstate refinements, rerun the same annual cohort, and address remaining voltage, reactive and rating violations separately from flow fit.

Current delivery: reproducible AC model, interactive map, source registers and handoff package. The latest wind/operator findings are diagnostic; they have not yet been promoted as a new repair.

Source: [4,6] Current delivery manifest and post-repair diagnosis. Source hashes and editable plot data accompany this deck.

Appendix: Counts, Units and Comparison Scope

Representation	Meaning
57 buses / 94 AC branches	Released 2019 benchmark; 271 generators, four DC records
140 buses / 225 AC rows	Derived unreduced counterpart; separate from the release
191 station/terminal rows	140 original + 51 additions; eight added ports retained
266 / 267 detailed AC buses	191 + 75 section points + one dated Dover regulated port
1,466 generator rows	1,437 native + 27 regional + HQ + PV20 equivalents
Six transfer records	Four inherited + two Ontario PAR equivalents

$$\mathrm{MAE} = \frac{1}{N}\sum_t |\hat{F}_t - F_t|, \qquad \mathrm{WAPE} = 100\frac{\sum_t |\hat{F}_t - F_t|}{\sum_t |F_t|}.$$

Public MW = model MW/γ, $\gamma = 0.3001207518$; amperes are never divided by γ. The paper uses a different, limit-normalized metric. Cross-year error scores do not isolate topology effects.

Source: [1,2,4] Paper raw branch-count/source-version discrepancy is retained in the release audit; counts refer to named representations, not interchangeable networks.

Appendix: Public Generation Data and Assumptions

Source	Available information	What it does not establish
NYISO P63	Statewide output by fuel, nominal five-minute records	Named plant dispatch or geography
NYISO Gold Book	Named fleet, capability and prior-year annual energy	Current-year hourly telemetry
EIA-923	Plant energy; reporting frequency from source flags	Intra-month output and curtailment
NYISO P27	Delayed masked offers and commitment parameters	Metered output; Dispatch MW fields are bid points
P34 / EIA-930	Reported device flow / regional interchange	Exact circuit-meter correspondence

Assumptions: bus load shares, hourly plant allocation, capability bounds, equivalent connections, reactive policy and meter approximations. EIA flag AM means monthly values reported annually; A means annual only.

Source: [3,6,7] Official product schemas and EIA instructions. The verified 2025 Gold Book reports 2024 annual generation.

Appendix: Sources and Reproducibility

1. Liu, Yuan, Wang, Sward, Zhang and Anderson. *An Open Source Representation for the NYS Electric Grid to Support Power Grid and Market Transition Studies*. IEEE TPWRS 38(4), 2023. doi:10.1109/TPWRS.2022.3200887. Pinned author release and 2019-input audit.
2. Downstate/slim registers: bus and branch identities, replaced aggregates, retained ports and source-derived figure data.
3. Input, western-source, line-parameter and Dover-control audits; NYISO and EIA source receipts.
4. Current model and annual run `moses_repair_v1`; independent input, AC physics, recovery and scoring checks.
5. Parent 41 and 185/195 physical-repair report; current parameter and compensation provenance.
6. Post-repair diagnosis: annual ablations, wind audit and hourly-shape checks. Research snapshot `0e73b46`.
7. NYISO 2025 operating-study definitions; real-time fuel mix; 2025 Gold Book; EIA-923 instructions.

Companion sources: `source_manifest.json` contains file hashes; `data/` contains exact chart values; `presenter_notes.md` expands the source and interpretation notes.